



Workflow Pipeline: PhysioNet MIT-BIH $\rightarrow$ Morlet Wavelet Scalograms (32 Scales) $\rightarrow$ CNN Beat Encoder $\rightarrow$ Conformer Sequence Block $\rightarrow$ Failure-Aware Diagnostic Readout

Core Takeaway for AI Cardiology: Strict record-level validation exposes critical arrhythmia triage failures that standard random splits conceal.